

Advik Sikligar

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Education

B.Sc. University of Illinois Chicago, B.Sc. in Biomedical Engineering – Chicago, IL

Sept 2022 – Dec 2026

Experience

Biological Microsystems Lab, BioMicrofluidics Research Assistant

Apr 2023 – present

- Developed software and hardware to streamline workflows for microfluidic devices
- Conducted maintenance and calibration of SLA, FDM, and microfabrication equipment
- Delivered weekly progress updates and data summaries during lab team meetings

Engineering Design Team @ UIC, President

Aug 2022 – Apr 2025

- Mechanical and Electrical Lead for the UAV team leading the design, CAD, simulation, and testing
- Acted as a liaison between the team and university administration
- Balanced budget across 4 sub-teams while assisting with sponsorships, and outreach
- Coordinated logistics, meetings, and onboarding for new members

Projects

Cat's Eye

July 2025 – May 2026

Capstone project developing a portable bulbar conjunctiva imaging platform and computer vision tool for microvascular isolation and analysis.

- Built automated Python computer vision pipeline for vessel segmentation and morphological analysis
- Awarded Best in Show (Biomedical Engineering Division) at UIC Engineering Expo 2026

Tapioca

Jan 2023

Automated droplet segmentation via visual transformer

- Utilized SOTA ViT-based architecture for droplet segmentation and tracking
- Developed a easy-to-use Python package and front-end GUI
- Reduced analysis time from hours to minutes for rapid iteration and optimization

Publications

A systematic investigation of endothelial cell behavior under hydrostatic pressure

Oct 2025

Giulia Venturini, Advik Sikligar, Giulia De Campo, David T Eddington
10.1038/s41598-025-16094-2 (Scientific Reports)

Microfluidic Platform for Personalized Immunotherapy Screening with Core-Needle Biopsies

Oct 2025

Advik Sikligar, Yoo, Jeonghun, Lee, Steve, Eddington, David
(BMES Annual Meeting)

Skills

Languages: Python, C++ , VHDL, MATLAB, SQL, Bash/Shell

ML & Computer Vision: PyTorch, OpenCV, ONNX, NumPy, Pandas, Scikit-learn, CUDA

Embedded & Hardware: Arduino, Microcontrollers, VHDL, FPGA (Vivado, ISE)

Prototyping: Linux, Docker, Git, CI/CD

Awards

UIC Engineering Expo - Best in Show

May 2026

- Conjunctiva Bulbar Handheld Imaging Device (Biomedical Engineering Department)